Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “period” and “ellipse” to explain your model or result. Sentence starters are optional.

period: time for one complete cycle or orbit

ellipse: an oval-shaped closed curve with two focal points

orbit: a path around another body under gravity

01 Warm-up: bring back an earlier idea

Launch at 30 m/s from level ground at 25° or 65°.
Use g=10 and no drag.
Compare ranges and flight times.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

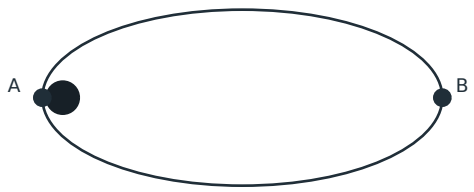
02 Worked example: explain the model

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

Worked reasoning: For an ideal Sun orbit, a=4 AU gives $T=\sqrt{(4^3)}=8$ years. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

Your turn: Draw an ellipse, put the Sun at a focus and mark the faster and slower parts.

Sentence starter: The ... in my model represents ... . Its label shows ... .

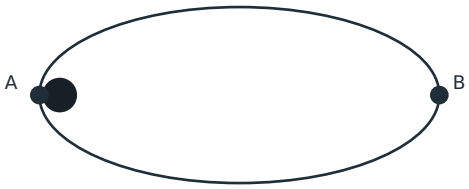


Marzano check - circle: ① ② ③

03 Read and interpret the model

Mark the Sun at a focus on an ellipse.
Label where the object is fastest and slowest.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A Sun-orbit has a=1.42 AU. Compute period. Then find period at a=4 AU and explain why doubling a does not double T.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Semimajor axis (AU)	Period (years)
1.42	____
4	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? What else is needed?

Sentence starter: I notice … . This supports … because … .

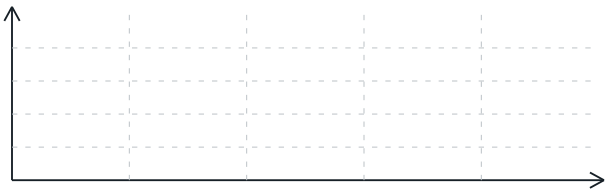
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A ball begins with downward velocity 4 m/s and falls 3 s.
Use downward positive, g=10 m/s² and no drag.
Find final velocity and displacement.
Sketch the motion and label a positive direction.

Sentence starter: I found … [unit]. This value means … in this situation.

Velocity (m/s)



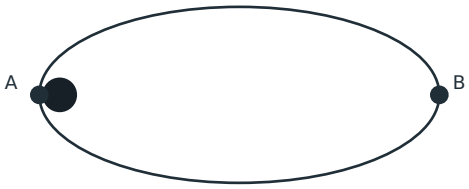
Time (s)

Marzano check - circle: 1 2 3

07 Exit check

For two Sun-orbits, $a_B=2\,a_A$.
Find T_B/T_A.
State why the same unit form cannot automatically be used for a Moon orbit about Earth.

Sentence starter: Under the assumption …, the evidence … supports … because … .



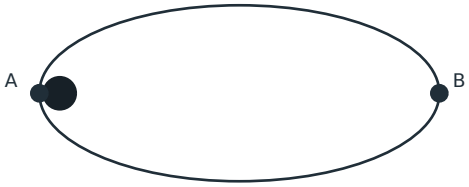
Marzano check - circle: 1 2 3

08 STRETCH: test the model

An ideal Sun orbit has period 27 years.
Find its semimajor axis. If eccentricity is 0.5, find perihelion and aphelion distances using $r=a(1\mp e)$.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3